Week 2&3 - Questions for Computer Vision

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Easy

1. **Who invented LeNet and what was its primary application?**
2. **What dataset was LeNet trained on?**
3. **What was AlexNet's biggest achievement in 2012?**
4. **Name two innovations introduced in AlexNet.**
5. **What activation function replaced sigmoid/tanh in AlexNet?**
6. **What does ReLU do to negative input values?**
7. **What is the purpose of dropout in training neural networks?**
8. **How many layers does VGG typically have?**
9. **What filter size did VGG standardize?**
10. **What problem does ResNet solve?**
11. **What is the key innovation in ResNet?**
12. **What does EfficientNet's compound scaling balance?**
13. **Which CNN architecture enabled training of networks with 152 layers?**
14. **Which architecture is lightweight yet accurate for mobile vision apps?**
15. **What does object detection aim to identify in an image?**
16. **What does R-CNN stand for?**
17. **Why is YOLO considered fast?**
18. **What does SSD stand for?**
19. **What is semantic segmentation?**
20. **What is the role of the encoder in U-Net?**

Medium

1. **How did LeNet prove CNNs were better than traditional ML methods?**
2. **Why was GPU training critical for AlexNet's success?**
3. **Compare accuracy before and after AlexNet on ImageNet.**

4. **Why does stacking small filters (like 3×3 in VGG) improve performance?**
5. **How do residual connections help gradient flow in ResNet?**
6. **What is the difference between direct mappings and residuals?**
7. **Why is EfficientNet more efficient than simply making networks deeper or wider?**
8. **Give one real-world application of ResNet.**
9. **Give one real-world application of EfficientNet.**
10. **Why is R-CNN slow compared to YOLO?**
11. **How does YOLO treat detection differently from R-CNN?**
12. **Why is SSD better at detecting objects of different sizes?**
13. **Which detection architecture would you choose for medical imaging and why?**
14. **Which detection architecture would you choose for autonomous driving and why?**
15. **What challenge does semantic segmentation face compared to classification?**
16. **How does FCN differ from traditional CNNs in its output?**
17. **Why are FCN predictions often coarse?**
18. **How do skip connections in U-Net improve segmentation accuracy?**
19. **What is the role of the decoder in U-Net?**
20. **Why is U-Net widely used in medical imaging?**

Hard

1. **Explain how dropout improves generalization with an example.**
2. **Why did AlexNet's use of ReLU help reduce the vanishing gradient problem?**
3. **How does compound scaling in EfficientNet systematically balance depth, width, and resolution?**
4. **Why do residual connections allow training of very deep networks without degradation?**

5. **Compare the trade-off between accuracy and speed in R-CNN, YOLO, and SSD.** *R-*
6. **How would you adapt YOLO for detecting very small objects?**
7. **Why are skip connections critical in both ResNet and U-Net, though used differently?**
8. **How does the encoder-decoder structure in U-Net ensure both context and precision?**
9. **If you were designing a segmentation model for satellite imagery, which architecture would you choose and why?**
10. **How did the evolution from LeNet → AlexNet → VGG → ResNet → EfficientNet shape the foundation for Vision Transformers?**

Mask R-CNN

1. What key limitation of traditional object detection does Mask R-CNN address?
2. How does Mask R-CNN extend Faster R-CNN?
3. What role does the Region Proposal Network (RPN) play in Mask R-CNN?
4. Why was RoI Align introduced, and how does it improve accuracy compared to RoI Pooling?
5. What are the three outputs produced by Mask R-CNN for each detected object?
6. Why are pixel-level masks more useful than bounding boxes in medical imaging?
7. Give two real-world applications of Mask R-CNN outside of medical imaging.
8. How does Mask R-CNN handle overlapping objects in crowded scenes?

Sequence Modeling for Video (CNN + LSTM)

9. Why can't CNNs alone capture temporal dependencies in video data?
10. What specific problem do LSTMs solve compared to vanilla RNNs?
11. Explain the role of the input, forget, and output gates in LSTMs.
12. In a CNN + LSTM pipeline, what does each component contribute to video understanding?
13. Provide an example of an application where CNN + LSTM is essential.
14. How does sequence modeling enable gesture recognition compared to static image classification?

RNNs vs LSTMs

15. What is the vanishing gradient problem, and why does it affect RNNs more than LSTMs?
16. Compare RNNs and LSTMs in terms of memory and best use cases.
17. Why are LSTMs better suited for variable-length sequences like video?
18. Give an example of a task where RNNs might still be sufficient instead of LSTMs.

Vision Transformers (ViT) & Attention

19. What is the purpose of the Query, Key, and Value in the attention mechanism?
20. Why does splitting an image into patches help Vision Transformers capture global context?